
Nirvana 2026

PeaceOfCode Hackathon



Track 2: Early Detection of Neurological Disorders

Background

Neurological disorders such as Alzheimer’s Disease (AD) & Mild Cognitive Impairment (MCI) often remain undiagnosed during their early stages due to subtle, heterogeneous, & nonspecific clinical manifestations. This delay in diagnosis significantly limits the effectiveness of therapeutic interventions & contributes to increased long-term healthcare costs & societal burden.

Magnetic Resonance Imaging (MRI) plays a vital role in identifying structural brain alterations associated with neurodegenerative diseases. However, the manual interpretation of MRI scans is time-consuming, demands specialized radiological expertise, & is often unavailable in primary care or resource-constrained healthcare settings.

Recent advancements in deep learning & medical image analysis have enabled the development of automated, software-based screening systems capable of analyzing MRI data alongside associated clinical information. These AI-driven approaches facilitate early diagnostic assessment & disease characterization, while supporting non-specialist healthcare providers through reliable, scalable, & cost-effective decision-support solutions.

Problem Description

The objective of this project is to design & develop an AI-driven, web-based screening system for the early detection, classification, & assessment of neurological disorders using T1-weighted Magnetic Resonance Imaging (MRI) data in conjunction with CSV-based clinical class labels.

The provided dataset comprises:

- **T1-weighted MRI brain scans** of individual subjects
- **A CSV file** containing subject-level diagnostic information that categorizes neurological status into clinically relevant classes, including:
 - Cognitively Normal (CN)
 - Mild Cognitive Impairment (MCI)
 - Alzheimer’s Disease (AD)

The proposed system must function entirely as a **software-only, web-based solution**, without reliance on custom hardware, wearable sensors, or physical diagnostic kits. The final application should be lightweight, scalable, & easy to deploy, enabling its use in both clinical & field settings. Furthermore, the system should be designed for accessibility by **non-specialist healthcare workers**, providing intuitive decision support to assist in early screening & referral for further clinical evaluation.

Dataset Link

[Download Dataset](#)

TASK 1 DATASET PREPROCESSING

Objective:

Prepare the provided MRI dataset for deep learning-based classification by performing standardized preprocessing & data organization steps.

Input:

- Raw T1-weighted MRI brain scans
- A CSV file containing subject-level class labels (CN, MCI, AD)

Scope / Operations:

- Organize MRI scans based on unique subject identifiers
- Map each MRI scan to its corresponding diagnostic label using the CSV file
- **Apply standard MRI preprocessing techniques, including but not limited to:**
 - Skull stripping to remove non-brain tissues
 - Spatial normalization/registration to a common anatomical space
 - Intensity normalization to reduce scanner-related variations
 - Resizing or resampling to ensure uniform input dimensions
- Split the processed dataset into **training**, **validation**, & **testing** subsets
- Ensure data consistency

Output:

A clean, standardized, & well-organized MRI dataset ready for binary & multi-class classification tasks

TASK 2 BINARY NEUROLOGICAL CONDITION CLASSIFICATION (CN VS AD)

Goal:

Build a deep learning-based medical imaging model to evaluate how accurately Alzheimer's Disease can be identified using T1-weighted MRI scans alone.

Objective:

Classify each subject into one of two neurological categories:

- Cognitively Normal (CN)
- Alzheimer's Disease (AD)

The objective is to assess whether structural brain information from MRI is sufficient to distinguish healthy individuals from those with Alzheimer's Disease.

Input:

- Preprocessed T1-weighted MRI brain scans
- Binary class labels (CN, AD) extracted from the provided CSV file

Allowed Features:

- MRI-derived spatial & intensity features
- Deep feature representations learned using CNN-based architectures (e.g., 2D/3D CNNs, ResNet-based backbones)

Output:

- A binary prediction indicating **Cognitively Normal** or **Alzheimer's Disease** for each subject
- **Threshold Accepted ≥91%**

Evaluation Metrics:

- Balanced Accuracy
- Area Under the ROC Curve (AUC)
- Macro F1-Score
- Precision & Recall (per class)
- Training & Validation Loss & Accuracy Curves
- Confusion Matrix

TASK 3 MULTI-CLASS NEUROLOGICAL STATE CLASSIFICATION (CN VS MCI VS AD)

Goal:

Build a deep learning-based medical imaging model to evaluate how well different stages of cognitive impairment can be distinguished using T1-weighted MRI scans.

Objective:

Classify each subject into one of the following neurological states:

- Cognitively Normal (CN)
- Mild Cognitive Impairment (MCI)
- Alzheimer's Disease (AD)

The objective is to assess the model's ability to capture subtle structural brain changes that characterize disease progression from normal cognition to mild impairment & advanced neurodegeneration.

Input:

- Preprocessed T1-weighted MRI brain scans
- Multi-class diagnostic labels (CN, MCI, AD) derived from the provided CSV file

Allowed Features:

- MRI-based spatial & volumetric features
- Deep feature representations learned using CNN-based architectures (e.g., 2D/3D CNNs, ResNet, or EfficientNet backbones)

Output:

- A multi-class prediction indicating the subject's neurological state: CN, MCI, or AD
- **Threshold Accepted >55%**

Evaluation Metrics:

- Balanced Accuracy
- Area Under the ROC Curve (AUC)
- Macro F1-Score
- Precision & Recall (per class)
- Training & Validation Loss & Accuracy curves
- Confusion Matrix

TASK 4 APPLICATION DEVELOPMENT & DEPLOYMENT

Goal:

Build a user-friendly web-based application that delivers an end-to-end AI-powered neurological screening solution using MRI data.

Objective:

Integrate the trained deep learning models into a single web-based platform that enables automated preprocessing, prediction, & visualization of neurological screening results for supporting clinical decisions.

Input:

- T1-weighted MRI brain scans uploaded by the user
- Optional subject-level metadata (linked via CSV format)

System Functionality / Features:

- MRI scan upload interface
- Automatic execution of the MRI preprocessing pipeline
- Option to select:
 - Binary classification (CN vs AD)
 - Multi-class classification (CN vs MCI vs AD)
- Real-time prediction using trained AI models
- Display of:
 - Predicted neurological class
 - Confidence/probability scores
- Intuitive & easy-to-use graphical interface designed for non-specialist healthcare workers

Output:

- Clear & interpretable screening results indicating neurological status
- Visual feedback supporting clinical referral & decision-making

FINAL DELIVERABLES

1. Standardized MRI preprocessing pipeline
2. Trained a binary classification model (CN vs AD)
3. Trained a multi-class classification model (CN vs MCI vs AD)
4. Comprehensive performance evaluation reports, including metrics & visualizations
5. Fully functional web-based neurological screening application

Good luck, and may your models achieve diagnostic excellence!